Jazzman's DIY Electrostatic Loudspeaker Page

DIY electrostatic loudspeakers

Jazzman's Ripol Subs:

The speaker building psychosis got the best of me again after becoming intrigued by the unique dipolar subwoofers patented by German speaker builder Axel Ridtahler. The name "RiPol" is short for "Ridtahler dipole".

I will share my CAD drawings and parts lists of the version shown here (rounded edges) or a simpler version with exposed plywood edges-- just email me: Jazzman1953@gmail.com

The woofers in a RiPol are in an opposing push/push configuration.

A RiPol is basically a compact, folded-baffle dipole, except the radiation pattern is not the classic figure-8 dipole but cardioid shaped, with a pronounced frontal lobe. It can play very low, and its cardioid pattern is less prone than monopole subs to excite the room's resonance which can give bass the dreaded "one-note boom".

A unique and beneficial RiPol characteristic is its effect on the woofers' fundamental resonance. Whereas a conventional sealed or ported box forces the woofer's resonance *higher* than it's resonance in free air, the Ripol's front chamber actually *lowers* the woofers' resonant frequency by as much as 10Hz (depending on chamber size, 7Hz is typical).

The result is lower distortion and clearer tonal quality because most music contains little or no energy that low, so the woofers' [lowered] resonance is is not excited or heard.

Ripoles pressurize the room differently than conventional subs.
I would describe their sound as quality over quantity-- clean and unobtrusive, arising from nowhere and receding back to nowhere.   Great for jazz-- not so much for HT sound effects.

The Peerless SLS 12" woofers are inexpensive and ideal for RiPol use. The US distributor is Parts Express, but are not always in stock. They are easier to find in Europe.

A commercial RiPol sub is available from Modal Akustik in Germany, and these use the very same Peerless SLS 12 woofers.

I found no published modeling data for a Ripol so I based the dimensions on guidelines posted by a respected Ripol guru who goes by the user name "Rudolf" on the DIY Audio Forum, which I've paraphrased below:

Area of front chamber opening should be 1/3 to 1/4 of woofers' combined piston area (SD).

For woofers having more than 10mm X-max, use 1/3 SD minimum.

For woofers with 10mm or less X-max, chamber area can be 1/4 SD.

Area of rear chamber opening should be 1/2 to 1 SD.

Chamber depth need not be greater than necessary to fit the woofer.

The front and rear chambers exhibit rather loud cavity resonances which must be either tamed with a notch filter (parametric EQ) or simply avoided by setting the crossover frequency at least one octave lower, using a 48db sloped filter, or one octave lower using a 24db filter.

For the Ripol shown here; the front chamber resonance peaks at 299Hz and the rear chambers at 294Hz. I use a digital crossover set low enough to avoid exciting the RiPol's cavity resonances.

The boxes are 3/4 red oak plywood, edge trimmed with quarter-round oak molding. The center C-section assembly glued up from solid white oak planks stained in a contrasting color for effect and indexed to the mating box halves with (4) 1/4 oak dowel pins on each side.

The assembly is held together with 1/4-20 all-thread rods and cap nuts as shown in the photos.

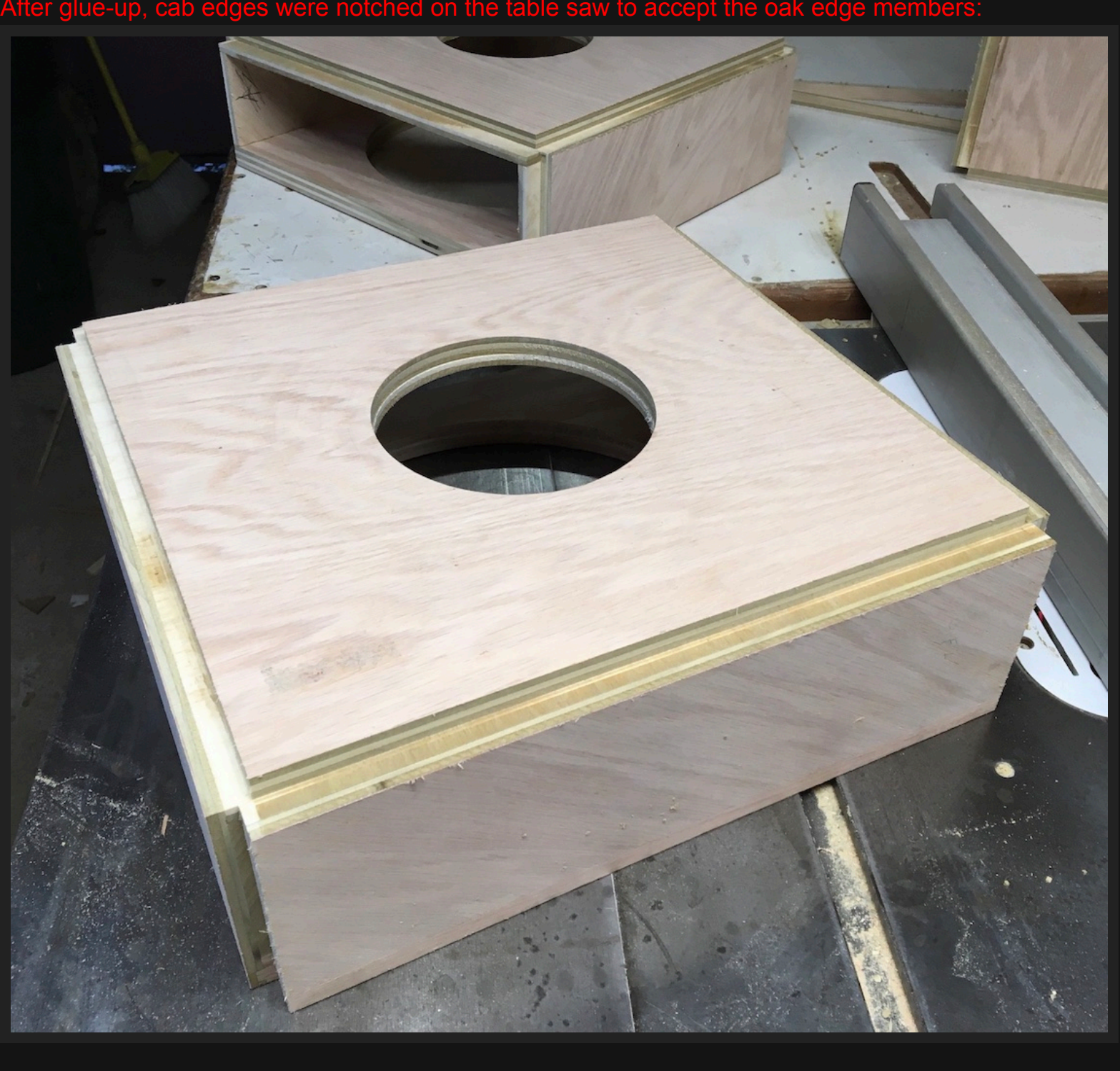
The woofer cutouts in the baffle boards are cut with a bit of extra clearance to allow the woofer magnets to center themselves in the holes in the outer panel. The holes in the outer panels allow .090" of clearance for the speaker magnets and I used foam speaker gasket tape to seal around the woofer magnets.

The two 8-Ohm woofers are wired in parallel and in phase (push/push, not push/pull), for 4-Ohm load

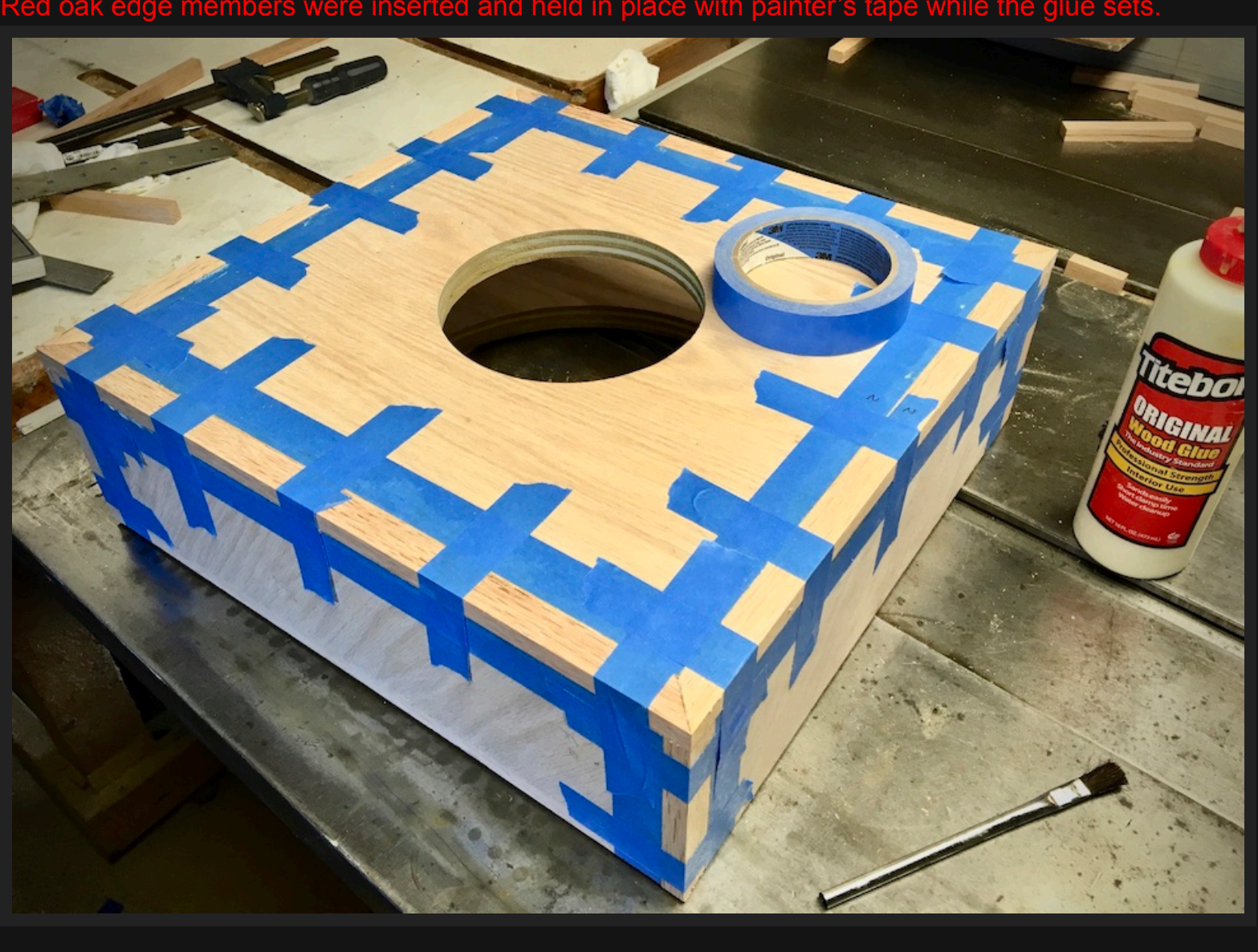
Rabbeted cab details bank together for gluing:

Below: Cab details clamped together while the glue sets.

After glue-up, cab edges were notched on the table saw to accept the oak edge members:

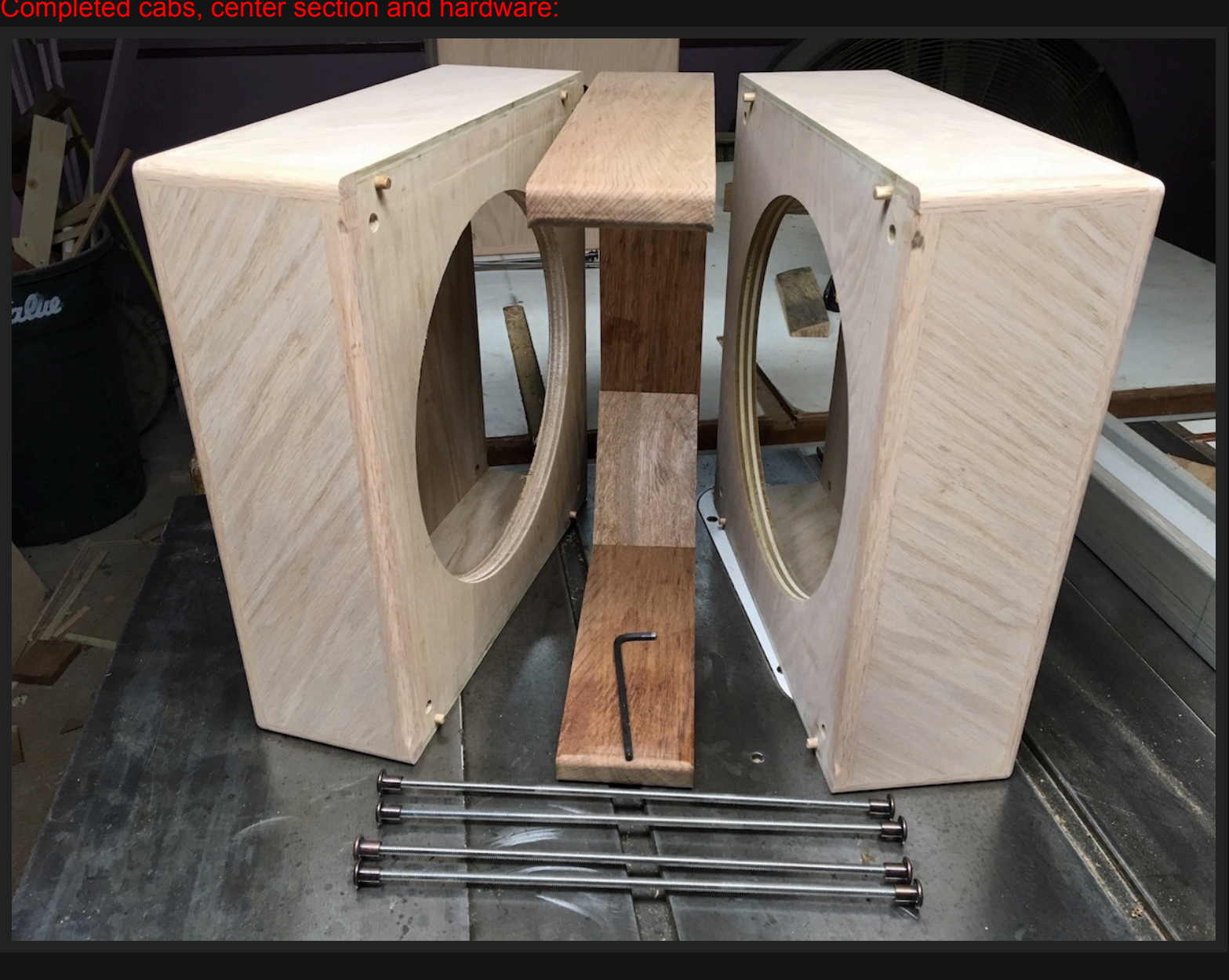


Red oak edge members were inserted and held in place with painter's tape while the glue sets.



Cab with mitered edge members glued in and sanded.

Completed cabs, center section and hardware:

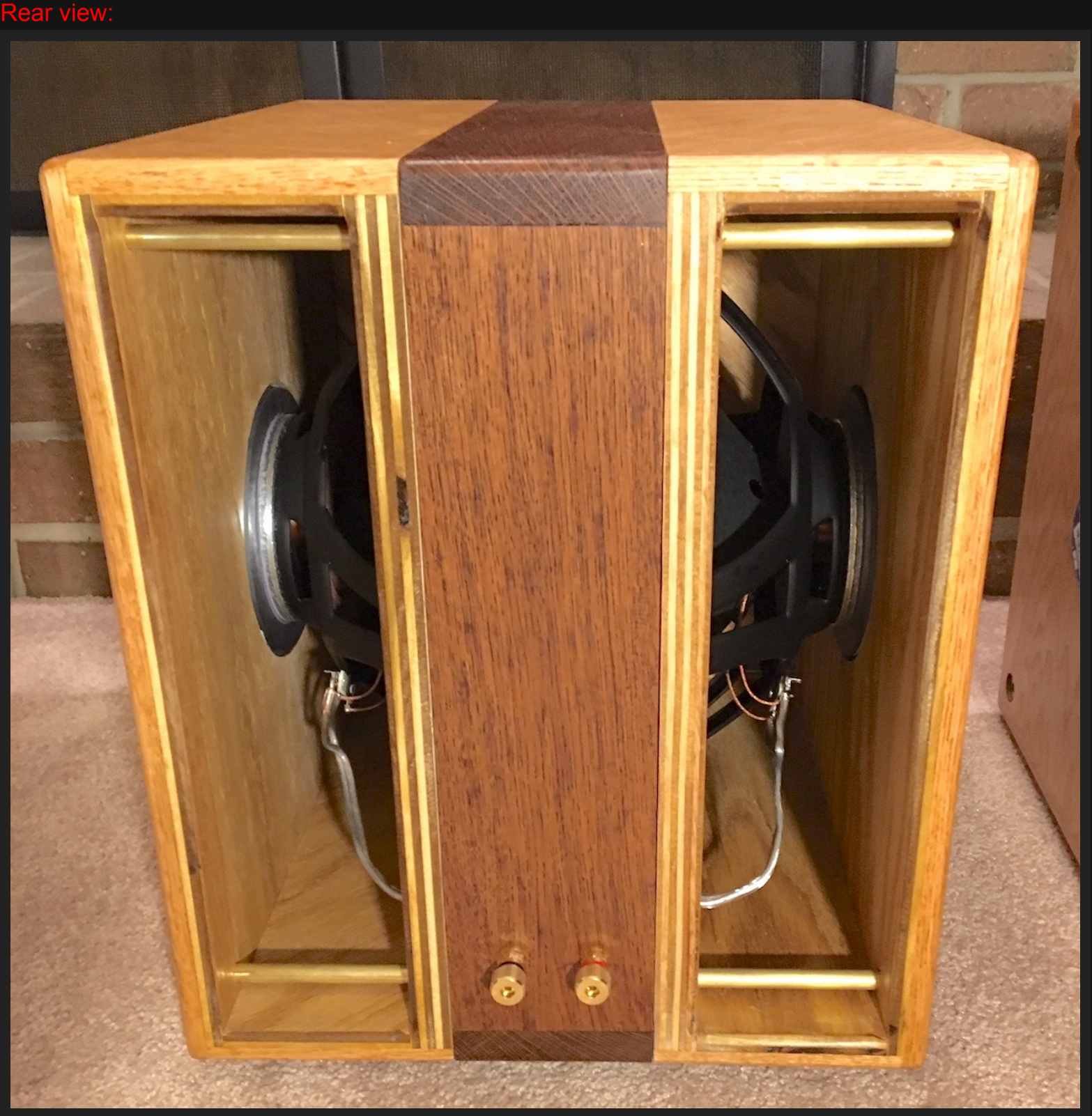


Subs were preassembled and edges were rounded-over to 3/8" radius on a router table:

Front view: Completed cabs ready for stain & satin clear coat.

Completed subs:

Rear view:



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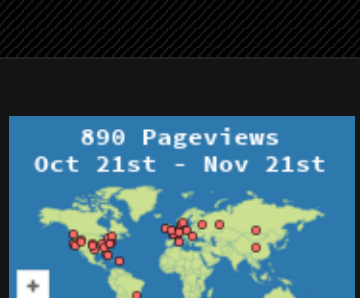

Jazzman's Stats

Hybrid Transmission Line ESL's

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